23/01/2026
Evening

Aakash
Medical | IIT-JEE | Foundations

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Memory Based
Answers & Solutions

Time : 3 hrs. *for* M.M. : 300

JEE (Main)-2026 (Online) Phase-1

(Physics, Chemistry and Mathematics)





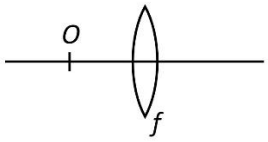
PHYSICS

SECTION - A

Multiple Choice Questions: This section contains 20 multiple choice questions. Each question has 4 choices (1), (2), (3) and (4), out of which **ONLY ONE** is correct.

Choose the correct answer :

1. When an object is kept at distance 8 cm and 24 cm from a convex lens magnitude of magnification is same in both cases. Find focal length of the lens.



(1) 32 cm (2) 8 cm

(3) 24 cm (4) 16 cm

Answer (4)

Sol. $m = m = \frac{f}{f+\mu}$

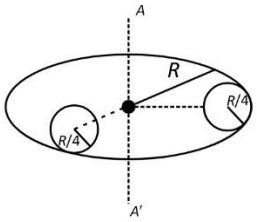
$$\frac{f}{f+8} - \left(\frac{f}{f-24}\right)$$

– $f + 24 = f - 8$

$32 = 2f$

$F = 16$ cm

2. From a uniform disk of radius R and mass M two small disk of radius $\frac{R}{4}$ is being cut as shown in figure. Find the moment if inertia of the system about exist AA′



(1) $\frac{79MR^2}{128}$ (2) $\frac{79MR^2}{256}$

(3) $\frac{109MR^2}{256}$ (4) $\frac{109MR^2}{128}$

Answer (3)

Sol. $\frac{MR^2}{2} - \left(\frac{MR^2}{16\times 2} + \frac{M9R^2}{16}\right)\times 2 = I$

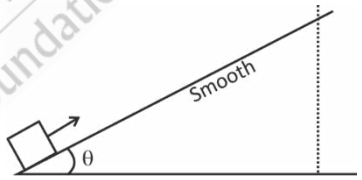
Also $m = \frac{M}{16}$

So $I = \frac{MR^2}{2} - \frac{2MR^2}{16}\left(\frac{1}{2}+9\right)$

$= \frac{MR^2}{2} - \frac{2MR^2}{16\times 16}\times\frac{19}{2}$

$= \frac{109}{256}MR^2$

3. A body is projected up the smooth incline plane having angle of inclination θ with the horizontal as shown in the figure. Find the distance covered before stopping.



(1) $\frac{u^2}{2g\cos\theta}$ (2) $\frac{u^2}{2g}$

(3) $\frac{u^2}{2g\sin\theta}$ (4) $\frac{u^2}{2g\tan\theta}$

Answer (3)

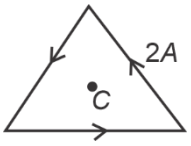
Sol. $mgh = \frac{1}{2}mu^2$

$\frac{u^2}{2g} = h$

$I = \frac{h}{\sin\theta} = \frac{u^2}{2g\sin\theta}$



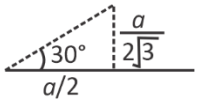
4. In equilateral triangular frame, then is current of 2*A*. The side of frame is $4\sqrt{3}$ cm. Magnetic field at center *C* is



(1) $30\sqrt{3}\ \mu T$ (2) $10\sqrt{3}\ \mu T$

(3) $3\sqrt{10}\ \mu T$ (4) $10\sqrt{10}\ \mu T$

Answer (1)

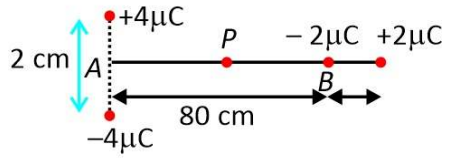
Sol. 

$$B=\frac{\mu_0 i\times 2\times C u30^\circ}{4\pi\left\{\frac{a}{2\sqrt{3}}\right\}}$$

$$B=\frac{\mu_0\sqrt{3}}{2\pi\times 2\times 10^{-2}}=10^{-5};$$

$$B_{net}=3\sqrt{3}\times 10^{-5}$$

5. Four charges are kept as shown in the figure. Find magnitude of electric field at point *P*. *P* is midpoint of line *AB*.

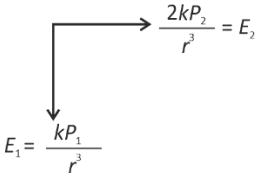


(1) 180 kV/m (2) $\frac{45\sqrt{5}}{8}$ kV/m

(3) 270 kV/m (4) $60\sqrt{3}$ kV/m

Answer (2)

Sol. $P_1=8\times 10^{-8}$ $P_2=2\times 10^{-8}$



$$E_1=\frac{9\times 10^9\times 8\times 10^{-8}}{64\times 10^{03}},\ E_2=\frac{9\times 10^9\times 2\times 2\times 10^{-8}}{64\times 10^{-3}}$$

$$=\frac{9}{8}\times 10^4 \qquad =\frac{9}{16}\times 10^4$$

$$E=\frac{9}{16}\sqrt{5}\times 10^4$$

$$=\frac{45\sqrt{5}}{8}\times 10^3$$

6. One mole of diatomic gas is expanding isothermally from *V* to 2*V* at 27°C. If the magnitude of work done by the gas in this case is same as the work done in adiabatic process where initial temperature is 27°C and final temperature is *T*°C. Find the value of *T*. [Use ln2 = 0.7]

(1) –37°C (2) –57°C

(3) –35°C (4) 0°C

Answer (2)

Sol. $\Delta w_1 = NRT\ell n(2) = 300R\ell n(2)$

$$\Delta w_2=N\frac{5}{2}R(T_i-T)$$

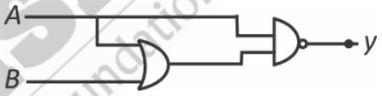
$$\Rightarrow \frac{5}{2}R(300-T)=300R\ell n(2)$$

$\Rightarrow$ $300 - T = 120\ell n(2)$ $[\ell n2 = 0.7]$

$\Rightarrow$ $300 - 84 = T = 216$ K

$\Rightarrow$ $T = -57°C$

7. Find the truth table for the given circuit.



(1)

A	B	Y
0	0	1
0	1	1
1	0	0
1	1	0

(2)

A	B	Y
0	0	0
0	1	1
1	0	1
1	1	1

(3)

A	B	Y
0	0	1
0	1	0
1	0	1
1	1	0

(4)

A	B	Y
0	0	1
0	1	0
1	0	0
1	1	1

Answer (1)

Sol. $\overline{A\cdot(A+B)}$

$\bar{A}+\overline{(A+B)}$

$\bar{A}+\bar{A}\cdot\bar{B}=\bar{A}$



8. An air bubble is moving upward from the bottom of lake having temperature 17°C. At the top the temperature of lake (and bubble) is 27°C. Assume no significant change in the density of the lake water upto its depth of 5m. Find the ratio of volume at top to bottom of the bubble.

(1) 1.25 (2) 1.75

(3) 1.55 (4) 1.85

Answer (3)

Sol. $P_{atm} = 10^5$ N/m^2 $\rho_\omega = 10^3$ kg/m^3

$g = 10$ m/s^2

$$\frac{10^5}{300} \times v_T = \frac{10^5 + 5 \times 10^3 \times g \times v_B}{290}$$

$$\frac{V_T}{V_B} = 1.5 \times \frac{30}{29} = 1.55$$

9. For a medium permeability $\mu = 2\mu_0$ and dielectric constant is 3. Find the ratio of speed in vacuum to speed in medium.

(1) $\sqrt{3}$ (2) 3

(3) $\sqrt{6}$ (4) 6

Answer (3)

Sol. $$v = \frac{1}{\sqrt{\mu\varepsilon}}$$

So $$\frac{c}{v} = \sqrt{\frac{\mu\varepsilon}{\mu_0\varepsilon_0}}$$

$$\frac{c}{v} = \sqrt{6}$$

10. EMF of two cells are measured using potentiometer method. If the balance lengths are 200 cm and 150 cm respectively. If the least count is 1 cm then find % error in calculating $\frac{E_1}{E_2}$.

(1) 1.2% (2) 1.16%

(3) 0.50% (4) 0.75%

Answer (2)

Sol. $$E_1 = \left(\frac{E_0}{l_0}\right) l_1$$

$$E_2 = \left(\frac{E_0}{l_0}\right) l_2$$

$$\frac{E_1}{E_2} = \frac{l_1}{l_2}$$

So, $$\left(\frac{\Delta l_1}{l_1} + \frac{\Delta l_2}{l_2}\right) \times 100 \approx 1.16\ \%$$

11. A man jumps from a plane, after 2 seconds he open parachute due to which if he retarded with 3 m/s^2. When the man is at 10 m height from ground his speed is 5 m/s. Find height of the plane when he jumped.

(1) 92.5 m (2) 90 m

(3) 85 m (4) 110 m

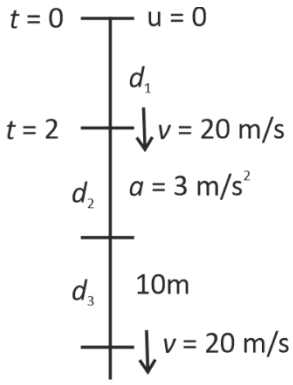
Answer (1)

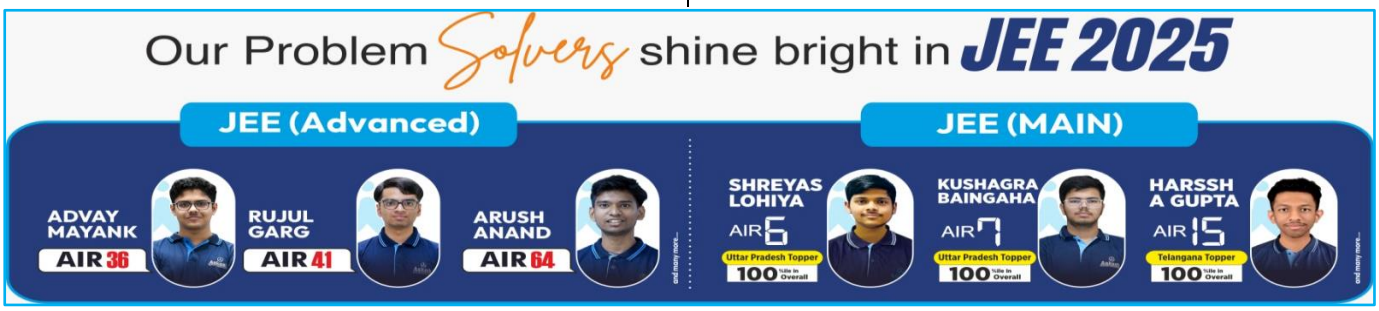
Sol. $d_1 = \frac{1}{2} \times 10 \times 4 = 20$ m

$$d_2 = \frac{400 - 25}{2 \times 3} = \frac{375}{6} = 62.5$$

$d = 20 + 62.5 + 10$

$= 92.5$





12. 

In a long solenoid of cross-section radius of 2 cm and of 500/cm turns density. A ring moves with constant speed 10 cm/s with axis coinciding with axis of solenoid. The radius and resistance of ring is 1 cm and 10Ω. Find heat dissipated in ring while it transverses 10 cm of distance. The current in solenoid is $I = 10\cos(100\pi t)$

(1) 300 μJ (2) 200 μJ

(3) 700 μJ (4) 850 μJ

Answer (2)

Sol. $\phi = BA$

$\phi = \mu_0 niA$

$$\varepsilon = -\mu_0 nA\frac{di}{dt} = \mu_0 nA\, i_0\omega\sin\omega t$$

$$\int Pdt = \int \frac{\mu_0^2 n^2 A^2 i_0^2 \omega^2}{R}\sin^2\omega t\, dt$$

$$= \frac{\mu_0^2 n^2 A^2 i_0^2 \omega^2}{R}\times\frac{1}{2}$$

$$= \frac{(4\pi\times10^{-7})^2\times(50000)^2(\pi\times10^{-4})^2 100\times10^4\times\pi^2}{10\times2}$$

$$= \frac{16}{2}\times10\times10^{-14}\times25\times10^8\times10\times10^{-8}\times10^6\times\frac{10}{10}$$

$$= \frac{400}{2}\times10^{1-14+8+1-8+6}$$

$$= \frac{400}{2}\times10^{-6}\text{ J}$$

$= 200 \times 10^{-6}$

13. A small spherical ball of diameter 2 mm & density 10.5 gm/cc is dropped into a large column of viscous liquid. The density of liquid is 1.5 gm/cc and coefficient of viscosity is 10 poise calculate the terminal velocity (g = 10 m/s^2).

(1) 1 cm/s (2) 2 cm/s

(3) 3.5 cm/s (4) 1.5 cm/s

Answer (2)

Sol. $V_t = \frac{2}{9}\frac{r^2 g}{\eta}(\sigma-\rho)$

$= 20 \times 10^{-3}$ m/s $= 2 \times 10^{-2}$ m/s

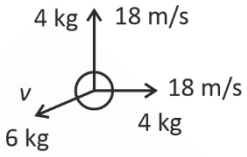
14. A mass of 14 kg is exploded into three fragments of 2 : 2 : 3 and both equal fragments fly off with same speed 18 m/s in mutually perpendicular direction. Then find the speed of the third fragment immediately after the explosion.

(1) $12\sqrt{2}$ (2) $6\sqrt{2}$

(3) $8\sqrt{2}$ (4) $10\sqrt{2}$

Answer (1)

Sol.



$\vec{P}_1 + \vec{P}_2 + \vec{P}_3 = 0$

$\vec{P}_1 + \vec{P}_2 = \vec{P}_3$

$\Rightarrow 6 \times v = \sqrt{2} \times 72$

$V = 12\sqrt{2}$ m/s

15. Speed of sound in air at 0°C is v. Then at what temperature (°C) the speed of sound becomes $2v$?

(1) 732°C (2) 1092°C

(3) 975°C (4) 819°C

Answer (4)

Sol. $v = \sqrt{CT}$

So $\frac{v}{v'} = \sqrt{\frac{T}{T'}}$

$\Rightarrow \frac{v}{v'} = \sqrt{\frac{273}{T}} = \frac{1}{2}$

$\Rightarrow T = 273 \times 4 = 1092$ K

So, $T = 819$°C



16. There is electric field in space given as $\vec{E}=\frac{A}{r^2}\hat{r}$ (A is constant). Thera are two point charges of –2 μC and 7 μC present at (9, 0, 0) and (–9, 0, 0) respectively. If electric potential energy of system is zero then A in SI units is

(1) $0.63\times10^{+3}$

(2) –0.325

(3) 1.26×10^{4}

(4) 0.325

Answer (3)

Sol. $\frac{Kq}{r^2}=\frac{A}{r^2}$

$q=\frac{A}{k}$

$U=\sum\frac{kq_iq_j}{r_{ij}}$

$U=\frac{K(-14)\times10^{-12}}{18}+K(5)\frac{A}{k\cdot g}$

$=\frac{9\times10^{9}\times10^{-12}}{18}+K(5)\frac{A}{k\cdot g}$

$U=-7\times10^{-3}+\frac{5A\times10^{-6}}{9}$

$10^{6}\times\frac{63}{5}\times10^{-3}=A$

$12.6\times10^{+3}=A$

17. Fission of single nucleus of U-235 liberates energy of 96 MeV. Energy released by fission of 47 gm of uranium is _____.

(1) 1.84×10^{12} J

(2) 3.28×10^{9} J

(3) 1.42×10^{15} J

(4) 3.21×10^{14} J

Answer (1)

Sol. Moles = $\frac{47}{235}=\frac{1}{5}$ mole

N = $\frac{6\times10^{23}}{5}$

E = $NE=\frac{6\times10^{23}}{5}\times96\times10^{6}\times1.6\times10^{-19}$

= $\frac{6\times96\times1.6}{5}=10^{10}$

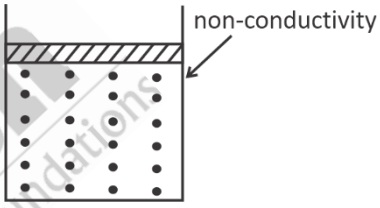
= 1.84×10^{12} J

18. 1 mole of gas is enclosed in a vertical cylinder and sealed with a massless frictionless piston as shown. On supplying 126 J of heat its temperature raised by 4°C. Find the height moved by piston due to the heat transfer. Internal energy of the gas is given by $U = 3nRT$

($P = 10^5$ N/m^2)

($R = 8.3$ J/K)

($A_{piston} = 17$ cm^2)



(1) 12.50 cm

(2) 13.50 cm

(3) 14.50 cm

(4) 15.50 cm

Answer (4)

Sol. $\Delta Q = \Delta U + P\Delta V$

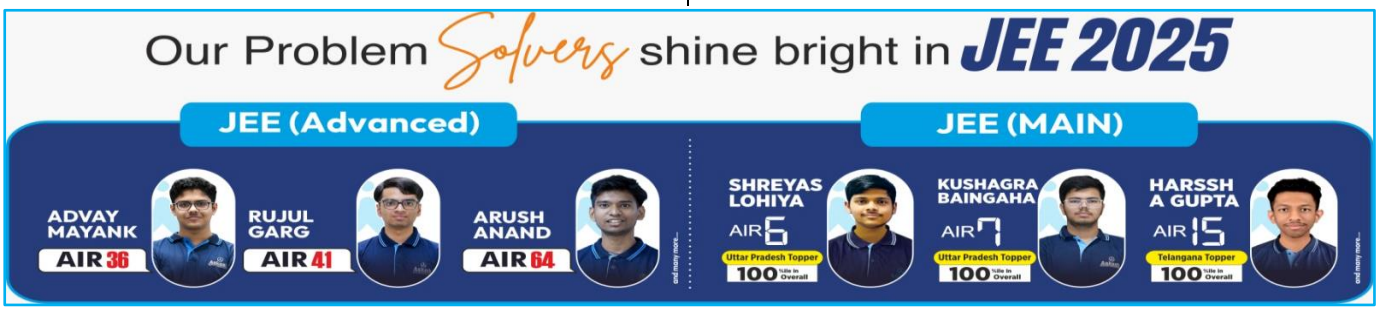
$126 = 3\times8.3\times4 + 10^5\times16\times10^{-4}\times h$

$126 = 99.6 + 1.7\,h\times10^2$

$h = 15.53$ cm

19.

20.



SECTION - B

Numerical Value Type Questions: This section contains 5 Numerical based questions. The answer to each question should be rounded-off to the nearest integer.

21. If r = radius of ball

ρ = density of liquid

σ = density of ball

t = time

A = constant

η = viscosity

For the expression $t = A(\rho)^a (\sigma)^b r^c (\eta)^d$

Then the value of $\frac{c-d}{a+b}$ is

Answer (3)

Sol. $T = \left(\frac{M}{L^3}\right)^{a+b} (L)^c (ML^{-1}T^{-1})^d$

Let $a + b = Z$

Then $Z + d = 0 \Rightarrow Z = -d$

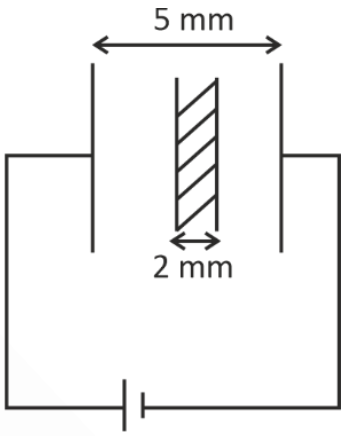
$c - 3Z - d = 0 \quad c - 2Z = 0 \Rightarrow c = 2z$

$-d = 1 ; Z = 1 = (a + b)$

$c = 2$

So, $\frac{c-d}{(a+b)} = \frac{2+1}{1} = 3$

22. An empty capacitor is charged with charge Q when attached with battery as shown. Now a mica sheet of 2 mm thickness is inserted and charge on capacitor $\frac{5}{4}a$ then the dielectric constant of mica is _____ .



Answer (2)

Sol. $\frac{Q_1}{Q_2} = \frac{C_1}{C_2} \quad \frac{Q}{\frac{5}{4}Q} = \frac{\frac{\varepsilon_0 A}{d}}{\frac{\varepsilon_0 A}{d - t + \frac{t}{K}}}$

$\Rightarrow \frac{4}{5} = \frac{3 + \frac{2}{K}}{5} \Rightarrow K = 2$

23.

24.

25.



CHEMISTRY

SECTION - A

Multiple Choice Questions: This section contains 20 multiple choice questions. Each question has 4 choices (1), (2), (3) and (4), out of which **ONLY ONE** is correct.

Choose the correct answer :

1. For XeO_2F_2, select the correct statement(s).

   (A) It shows see-saw shape.

   (B) Number of lone pair(s) of e^- on Xe is 1.

   (C) $\angle FXeF = 180°$ (approx.)

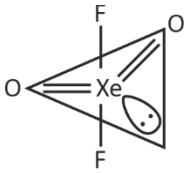
   (D) It has tetrahedral shape.

   (1) (A), (C), (D) Only (2) (A), (B) only

   (3) (A), (B), (C) only (4) (B), (C), (D) only

Answer (3)

Sol.



2. Statement I : Size of O^{2-} is smaller than F^-.

   Statement II : Electronegativity of F is more than that of oxygen.

   In the light of above statements, choose the correct option.

   (1) Both Statement I and Statement II are correct

   (2) Both Statement I and Statement II are incorrect

   (3) Statement I is correct but Statement II is incorrect

   (4) Statement I is incorrect but Statement II is correct

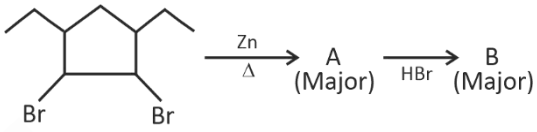
Answer (4)

Sol. Number of proton in O^{2-} is 8 and number of proton in F^- is 9 (both have $10e^-$). So, O^{2-} is larger than F^-

So, Statement I is false.

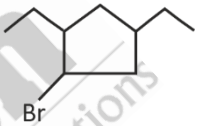
Statement II is true, F is smaller than O, so F is more electronegative than O.

3. Consider the reaction,



Choose the correct option,

(1) A is 

(2) B is 

(3) B is 

(4) A is 

Answer (3)

Sol.

